

7th Grade Unit 12 Assessment Guide Calculator

General Information	The questions on the unit assessment are divided into two parts to ensure questions for benchmarks MA.7.GR.2.1 and MA.7.GR.2.2 are assessed separately. The draft test item specifications state that questions for MA.7.GR.2.1 should not require students to use a formula, whereas the draft specifications state that questions for MA.7.GR.2.2 must include the formula for surface area. Essentially, questions for MA.7.GR.2.2 could clue questions for MA.7.GR.2.1 so the questions were separated. Teachers should set expectations on student work. Students who become proficient with the calculator could possibly enter the values and not show any work. A student who correctly enters the values using the formula into the calculator could very well have a complete understanding of the benchmark, so the student should not be punished for their capability. It may be best to state that the least amount of work you expect to see is what they entered into their calculator. This will allow for you to determine if a student had an accidental calculator entry problem.
Calculator Information	The questions on the unit assessment are divided into two parts - calculator and no calculator. The questions on the no calculator part are calculator neutral which means that a calculator does not give a student an advantage or disadvantage when showing their knowledge of the intended benchmark. All questions on the no calculator part asks students to leave their answer in pi form. The questions on the calculator part indicate the place value students should round to. Some of the questions indicate which value of pi to use. Teachers should determine how many, if any, points to deduct if a student does not use the correct value of pi. The draft item specifications state "Items that ask for an approximate answer must allow for the use of the exact value of pi (π) or any common approximation of pi (π)." This indicates that items on this assessment will allow students to use any common approximation. If the question is a multiple-choice question, then the student would choose the answer that is the closest to their answer if they use a common approximation other than the one given.
Total Recommended Points	The total recommended points for the no calculator portion of the assessment are 48 points. The total recommended points for the assessment are 100 points.

Question	1a	Benchmark(s)	MA.7.GR.2.2
	53,807.04 mm ²	Recommended Scoring (8 Total Points)	Consider giving students 5 points for work that shows correct use of the formula or net to determine the surface area. Since students are using a calculator, consider 2 points for the correct answer. Consider 1 point for the correct units.
		Additional Notes	The answer that is given is for using $\pi = 3.14$. As stated in the calculator information, the draft item specifications indicate that students can use any value of pi. It is left to the teacher to determine if accepting other answers is appropriate for their class.

Question	1b	Benchmark(s)	MA.7.GR.2.3
	955,162.9 mm ³	Recommended Scoring (8 Total Points)	Consider giving students 5 points for work that shows correct use of the formula determine the volume. Since students are using a calculator, consider 2 points for the correct answer. Consider 1 point for the correct units.
		Additional Notes	The answer that is given is for using $\pi = 3.14$. As stated in the calculator information, the draft item specifications indicate that students can use any value of pi. It is left to the teacher to determine if accepting other answers is appropriate for their class.

Question	2a	Benchmark(s)	MA.7.GR.2.2
	101.657 cm² or 101.605 cm² or 101.698 cm² Answers if students included the bamboo top 106.804 cm² or 106.847 cm² or 106.75 cm²	Recommended Scoring (8 Total Points)	Consider giving students 5 points for work that shows correct use of the formula or net to determine the surface area. Consider deducting 4 points if the student uses the diameter instead of the radius. Consider deducting 2 points if the student includes the bamboo top. Since students are using a calculator, consider 2 points for the correct answer. Consider 1 point for the correct units.
		Additional Notes	The answers provided use the value of pi on the calculator, $\pi = 3.14$, and $\pi = \frac{22}{7}$. If your students use another approximation for pi, then add that value to the possible answers.

Question	2b	Benchmark(s)	MA.7.GR.2.3
	61.77 cm³ or 61.74 cm³ or 61.79 cm³	Recommended Scoring (8 Total Points)	Consider giving students 5 points for work that shows correct use of the formula determine the volume. Consider deducting 4 points if the student uses the diameter instead of the radius. Since students are using a calculator consider 2 points for the correct answer. Consider 1 point for the correct units.
		Additional Notes	The answers provided use the value of pi on the calculator, $\pi = 3.14$, and $\pi = \frac{22}{7}$. If your students use another approximation for pi, then add that value to the answers.

Question	3	Benchmark(s)	MA.7.GR.2.3
	88.357 in.³ or 88.313 in.³ or 88.393 in.³	Recommended Scoring (8 Total Points)	Consider giving students 3 points for work that shows correct use of the formula to determine the volume of one piece of candy but do not find the total volume. Consider deducting 4 points if the student uses the diameter instead of the radius. Since students are using a calculator, consider 2 points for the correct answer. Consider 1 point for the correct units.
		Additional Notes	The answers provided use the value of pi on the calculator, $\pi = 3.14$, and $\pi = \frac{22}{7}$. If your students use another approximation for pi, then add that value to the answers.

Question	4	Benchmark(s)	MA.7.GR.2.3
7.40 in.	Recommended Scoring (8 Total Points)	Consider giving students 5 points for work that shows correct use of the volume formula to determine the height. Consider deducting 4 points if the student uses the diameter instead of the radius. Since students are using a calculator, consider 2 points for the correct answer. Consider 1 point for the correct units.	
	Additional Notes	The answer provided is the same for any of the common values of pi. If your students use another approximation for pi, then add that value to the answer.	